JACK MICHELINI

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EDUCATION

University of Michigan, Ann Arbor, MI
Bachelor of Science in Engineering, Computer Science *Dec 2025*
Coursework: Data Structures and Algorithms, Foundations of Computer Science, Introduction to Computer Organization, Linear Algebra, Introduction to Computer Security, Software Engineering, Web Systems, Election Cybersecurity, Mobile App Development for Entrepreneurs, Computational Linguistics

SKILLS

Languages: C, C++, Gherkin, JavaScript, Julia, MATLAB, Python, SQL, Bash
Other: Docker, Flask, Git, Github, Google Cloud Platform, Jinja, jQuery, Playwright, React.js, REST APIs, Salesforce

WORK EXPERIENCE

Intellious Technologies **May 2025 - Aug 2025**
Software Engineer Intern
- Served as a contractor for Commerce (formerly BigCommerce) to aid in integration testing during a large-scale migration to a newer Salesforce platform.
- Practiced test-driven development by designing end-to-end testing scenarios in Gherkin and executing them with Playwright to validate both UI and API elements, ensuring platform functionality and data integrity.
- Deployed testing scenarios to run fully-automated on CI pipeline in Github Actions in a regression testing set, streamlining testing and significantly reducing manual testing time.

Camp Michigania
Store Staff - Seasonal *May 2026 - Aug 2026*
- Served as main salesperson to sell thousands of dollars worth of merchandise to campers while prioritizing customer experience
- Restocked and managed inventory to ensure consistency with Salesforce inventory system.

Boating Staff - Seasonal *May 2022 - Aug 2023*
- Collaborated with a team performing maintenance on Camp Michigania's fleet of over 50 hand-powered boats, sailboats, and motorboats to optimize for camper and staff operation.
- Devised programming for kids ranging in age from three to seventeen years old to provide experience and foster comfortability on the water.
- Facilitated sailing lessons and hands-on instruction to teenager and adult campers to ensure self-proficiency and understanding of both sailing fundamentals and more advanced concepts.

PROJECT EXPERIENCE

Test Automation Project **Winter 2024**
- Used AmericanFuzzyLop (AFL) program to automatically generate test files to maximize coverage of a libpng's test suite.
- Similarly used the EvoSuite generational tool to generate unit tests for jsoup Java library.

Client-Side Dynamic Web Pages Application **Fall 2024**
- Developed Instagram clone "insta485" in Python using Flask and Jinja for backend functionality, including user authentication (login/logout), user following, post liking, infinite scrolling, and live page updates.
- Built frontend client with React and JavaScript to interact with the REST API for seamless data requests and dynamic content rendering.

MapReduce Search Engine **Fall 2024**
- Implemented a MapReduce pipeline in Hadoop to provide text-based analysis of documents and generate an inverted index for efficient search functionality.
- Developed an indexing and search server with a dynamic user interface to handle queries and display results.

Cryptography Project **Winter 2024**
- Designed a Python program to exploit weaknesses in SHA and MD5 algorithms, Cipher-Block Chaining (CBC) encryption, and RSA Signatures.
- Utilized methods such as length extension attacks, hash collisions, CBC padding oracle attack, and Bleichenbacher's attack to exploit weaknesses in a timely and energy-efficient manner.